

# Erdős Problem #143

EP-143

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

## ORIGINAL PROBLEM STATEMENT

Let  $A \subset (1, \infty)$  be a countably infinite set such that for all  $x, y \in A$  and integers  $k \geq 1$  we have  $|kx - y| \geq 1$ . Does this imply that  $A$  is sparse? In particular, does this imply that  $\sum_{x \in A} \frac{1}{x \log x} < \infty$  or  $\sum_{\substack{x < n \\ x \in A}} \frac{1}{x} = o(\log n)$ ?